

A S SHAIK HASAN

B.Tech ECM · 2nd Year

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EDUCATION

B.Tech – Electronics and Computer Engineering

Expected May 2028

Vellore Institute of Technology, Chennai · **CGPA: 8.76 / 10.0**

TECHNICAL SKILLS

Languages: Python, JavaScript, C/C++, SystemVerilog, Java.

AI / ML / Data: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, CNN, Gemini AI API.

Web & Backend: React.js, Node.js, FastAPI, REST APIs, PostgreSQL, Vector Embeddings, RAG Pipelines

Hardware & Embedded: Arduino, ESP32, FPGA, Sensors & Actuators, IoT Systems

CS Fundamentals: Data Structures & Algorithms, OOP, SDLC, Agile / CI-CD Workflows.

Tools: Git, Linux, Vercel

PROJECTS

SerenaAI – Mental Health Support Platform

React.js · Node.js · PostgreSQL · Gemini AI · Vector Embeddings · RAG

- ▶ Built a production-ready full-stack AI mental health platform across the complete SDLC, integrating Gemini AI for natural language therapy conversations.
- ▶ Engineered a RAG pipeline using text and vector embeddings with PostgreSQL as the vector store, enabling the AI to retrieve relevant context and generate grounded, accurate responses.
- ▶ Architected a RESTful Node.js backend with React frontend, handling user sessions, conversation history, and real-time AI interaction.

Environmental Sound Classification (MR-AFCNN)

Research · Deep Learning

PyTorch · TensorFlow · CNN · Mel Spectrogram · ESC-50 · UrbanSound8K

- ▶ Designed MR-AFCNN, a novel lightweight CNN (~3M params) for Environmental Sound Classification using multi-resolution depthwise convolutions and frequency-spatial attention tailored for mel-spectrograms.
- ▶ Achieved 84.2% mean accuracy on ESC-50 (peak fold: 90%) via 5-fold cross-validation with SupCon loss, SWA, and SpecAugment — surpassing the human baseline (81.3%) and matching heavier SOTA CNNs.

ML-Based Smart Lock System

ESP32

ESP32 · DeepFace · OpenCV · Face Recognition

- ▶ Built a face recognition-based access control system using DeepFace and OpenCV for identity verification.
- ▶ Integrated the recognition pipeline with ESP32 to trigger lock/unlock mechanisms based on authenticated face matches.

Elevator Control System

SystemVerilog · FPGA

SystemVerilog · Digital Design · FSM

- ▶ Designed a multi-floor elevator controller in SystemVerilog using finite state machines, handling floor requests, door logic, and direction control.
- ▶ Simulated and verified the design with testbenches, ensuring correct behavior under edge cases and concurrent floor requests.

Emotion and Activity Sensitive RGB lighting system

Arduino · Embedded

YOLOv5 · FER Model · Computer Vision · FastAPI · React.js

- ▶ Built a real-time system using YOLOv5 for activity recognition and a FER model for facial emotion detection, streaming live inference data to a FastAPI backend.

- Mapped detection outputs to dynamic RGB lighting profiles via Arduino-controlled LEDs; built a React.js analytics dashboard displaying emotion logs, activity stats, and a patient stability monitor powered by the FastAPI backend.

LEADERSHIP & CLUB ACTIVITY

Web Developer – Biosphere, Vellore Institute of Technology, Chennai

2024 – Present

- Designed and developed the club's official website, owning both UI/UX design and full-stack implementation, deployed on Vercel.
- Organized and conducted a club hackathon with 100+ participants, managing event logistics, problem statements, and judging.

CERTIFICATIONS & COURSES

Machine Learning – [Coursera / Andrew Ng] · 2026

Python with Data Science – [NPTEL] · 2025